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POSTER

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When Context Grows, So Does the Challenge: Human Oversight in LLM Evaluation of Financial Tables

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Abstract

We evaluate the reliability of Large Language Models (LLMs)-as-a-Judge for comparing financial tables derived from text-to-SQL pipelines. Our findings show degraded performance on larger tables. To address this, we propose a triggered oversight framework that defers ambiguous cases to human experts.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

LLM, HCI, Financial AI

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1 Introduction

The financial industry generates massive and continually growing volumes of data, ranging from daily transactions and market activity to customer records, which are increasingly processed by AI systems [7]. These systems often produce complex outputs, such as summaries, risk assessments, or structured tables. However, the foundation of any successful AI application lies in the quality of its data and the rigor of its preparation, both requiring careful evaluation for accuracy and reliability [8].

Despite its transformative potential, AI adoption in finance faces significant challenges. Systems must handle messy data, inconsistent formats, and ambiguities that often require human judgment.

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Traditional workflows are manual, error-prone, and costly, while financial oversight has historically focused on external governance and compliance. The growing use of large language models (LLMs) introduces new complexities, calling for internal governance strategies centered on accountability and rigorous evaluation [8, 12].

A key barrier is the gap between the complexity of AI systems and the expertise of end users. Many stakeholders, including decision-makers, regulators, and practitioners, lack a clear understanding of how these models work, their limitations, or when they might fail [3]. This asymmetry creates two major risks:

- **Overtrust:** Users may place undue confidence in AI outputs, overlooking potential inaccuracies or hallucinations [12].
- **Undermined Trust:** In contrast, an opaque AI system may provoke unwarranted skepticism, leading to the under-utilization of its capabilities.

In either case, the interpretability of AI outputs and their alignment with user expertise are essential to building trust and reducing the risk of misinformed decision making. Among AI systems, LLMs have become especially prominent, yet their outputs present unique verification challenges. LLM outputs can be difficult to rely on, particularly in cases where models generate confident but incorrect information (hallucinations). These challenges are compounded by the limited transparency of many widely used models (from OpenAI, Anthropic, Google), which are commercially available but lack full access to model internals and training data. This opacity limits the use of explainable AI techniques to LLMs, restricting our understanding of their behavior [9].

These limitations have prompted the exploration of alternative evaluation strategies that do not rely on internal model transparency. Often considered the gold standard for evaluation, human evaluation by subject matter experts is expensive and time-consuming, hence not scalable. Thus, a growing paradigm for evaluating LLM outputs is 'LLM-as-a-Judge'. This approach uses large language models as evaluators for complex tasks, leveraging their ability to process diverse data types and deliver scalable, consistent, and cost-effective judgments. It has gained traction as an alternative to traditional expert-driven evaluations, particularly due to its efficiency and scalability [4, 5]. By integrating LLMs into the evaluation loop, systems can automatically handle straightforward cases, allowing human experts to concentrate on more ambiguous or high-stakes queries, such as those involving nuanced reasoning.

Despite the growing popularity of the “LLM-as-a-Judge” paradigm, these systems exhibit some key limitations:

- LLMs exhibit biases such as leniency, verbosity, and are sensitive to prompt complexity and length [2, 11, 13].
- LLMs may exhibit strong performance on general-purpose tasks but often struggle in domain-specific settings, where specialized knowledge and human expertise are crucial [1, 10]. For example, in a nuanced scenario where two tables produce the same answer but utilize different formats and terminologies, an LLM may mistakenly misclassify them.

An increasingly important use case for LLM-as-a-Judge is the evaluation of tabular results produced by SQL queries. In financial applications, such tables capture critical information about transactions, market activity, and trends. As LLMs are integrated into workflows, an open question remains: Can these models reliably compare complex, real-world financial tables? In this work, we investigate the effectiveness of LLM-as-a-Judge, focusing on table comparison tasks that arise when evaluating finance-specific text-to-SQL pipelines. In this work, we answer the following research questions :

- (Q1) How does accuracy change as LLM judges scale to larger financial tabular datasets?
- (Q2) How do we initiate human evaluations to determine the optimal balance between high accuracy and low manual evaluation effort?

2 Experiments

To ensure trustworthy evaluation of our text-to-SQL outputs, we build dedicated test sets for each use case, consisting of user queries paired with SQL code that retrieve relevant data. Since each SQL query may generate a dynamic table and we maintain hundreds of such cases across many deployments, manual review at scale is infeasible. Our evaluation pipeline must therefore be as automated as possible while maintaining high reliability.

Our approach begins with rule-based filters, ranging from simple string matches to more complex heuristics, to automatically handle a large portion of cases. Remaining examples are passed to an LLM for automated judgment, comparing the generated SQL’s output table with that of a gold query.

However, evaluating LLM reliability on real-world table outputs is difficult, as these tables are dynamic and often lack ground-truth annotations. To address this, we construct a synthetic evaluation set that mimics the real task while providing full control over the ground truth. Specifically, we introduce controlled perturbations to a fixed number of table cells, modifying numeric values, dates, or strings at various positions (e.g., at the edges, in the middle, or clustered). This setup allows us to systematically assess LLM judgment under known conditions.

We treat this as a binary classification task: an output is considered incorrect (negative label) if any corruption is present. Figure 1 shows an example from our dataset alongside its perturbed version.

We hypothesize that task difficulty increases with table size, resembling a “needle in a haystack” scenario where relevant information is buried in large amounts of data. To test this, we generate 50 perturbed (negative) examples for every 5-row increment in table size. Using our best-performing prompt, we collect LLM predictions

and compute accuracy against the ground truth. This allows us to identify thresholds where LLM performance degrades, enabling manual review to be triggered only when necessary.

	companyname	revenue	dataitemname	unit	fiscalyear	periodtypename	currencyname	simpleindustrydescription
0	XNET Corporation	5547.0	Total Revenues	Millions	2024	Annual	Japanese Yen	Software
1	Kyosha Co., Ltd.	24580.0	Total Revenues	Millions	2024	Annual	Japanese Yen	Electronic Equipment, Instruments and Components
2	Tokyo Electron Device Limited	242888.0	Total Revenues	Millions	2024	Annual	Japanese Yen	Electronic Equipment, Instruments and Components
3	Hakuto Co., Ltd.	182046.0	Total Revenues	Millions	2024	Annual	Japanese Yen	Electronic Equipment, Instruments and Components
4	Enplas Corporation	37805.0	Total Revenues	Millions	2024	Annual	Japanese Yen	Electronic Equipment, Instruments and Components

	companyname	industrydescription	revenue	revenueunit	unit	fiscalyear	periodtypename	currencyname
0	XNET Corporation	Software	5547.0	Total Revenues	Millions	2024	Annual	Japanese Yen
1	Enplas Corporation	Electronic Equipment, Instruments and Components	37805.0	Total Revenues	Millions	2024	Annual	Japanese Yen
2	Hakuto Co., Ltd.	Electronic Equipment, Instruments and Components	182046.0	Total Revenues	Millions	2024	Annual	Japanese Yen
3	Kyosha Co., Ltd.	Electronic Equipment, Instruments and Components	24579.0	Total Revenues	Millions	2024	Annual	Japanese Yen
4	Tokyo Electron Device Limited	Electronic Equipment, Instruments and Components	242888.0	Total Revenues	Millions	2024	Annual	Japanese Yen

Figure 1: Dataset Example. Top : Original table. Bottom : Perturbed table with a single revenue value changed. Rows and columns are shuffled, and column headers are renamed – cosmetic changes that preserve the table’s semantics. The only meaningful modification is the revenue value.

Figure 2 shows that the performance of frontier models such as GPT-4o degrades as table size increases. This trend aligns with findings from [6]. As tables grow larger, models increasingly default to generic responses like “At a glance, the tables look the same,” indicating a superficial or heuristic-based evaluation rather than a careful comparison.

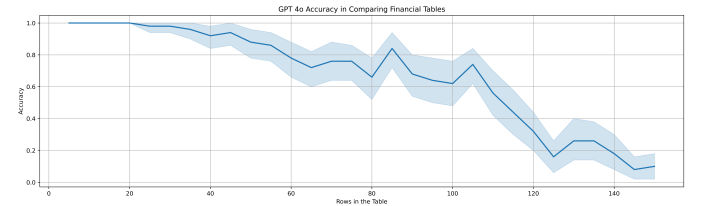


Figure 2: Performance of GPT-4o in Table Comparison Tasks. The model shows degraded performance as the tables get larger.

3 When Should Humans Step In

Our results in Section 2 indicate that LLMs become unreliable in table comparison tasks once the number of rows exceeds a threshold (60 in this case). However, manually reviewing all such cases is impractical at scale. To address this, we propose a set of decision criteria to determine when human intervention is necessary.

We begin by checking whether the two tables are structurally comparable, i.e., whether they have the same set of columns. However, structural equivalence is not always guaranteed. Columns may be renamed, tables may be transposed, or redundant information may be expressed in different forms. For example, one table might include a column with full currency names, while another uses

currency abbreviations. In many practical settings, either representation should be considered acceptable.

When the tables share a consistent structure, we align and sort them by a unique identifier (e.g., company name or transaction ID) and compare them in manageable chunks. If the tables are not directly comparable due to structural transformations or semantic ambiguity, we defer the evaluation to a human reviewer.

4 Conclusion

We conducted a comprehensive analysis of the capabilities of LLM-as-a-Judge in evaluating financial table comparison tasks. Our results show that as table size increases, model reliability significantly degrades, which poses risks in high-stakes real-world domains like finance. In response, we propose a set of criteria to determine when human oversight should be triggered during the evaluation process. This hybrid approach aims to balance scalability with accuracy, ensuring more trustworthy deployment of LLM-based systems in structured data pipelines.

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